

Soluzioni – Verifica di Goniometria

Esercizio 1 – Angoli notevoli (6 pt)

a)

$$\sin \frac{\pi}{6} = \frac{1}{2}$$

b)

$$\cos \frac{\pi}{3} = \frac{1}{2}$$

c)

$$\tan \frac{\pi}{4} = 1$$

d)

$$\sin \frac{\pi}{3} \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} = \frac{3}{4}$$

e)

$$2 \sin \frac{\pi}{6} - \cos \frac{\pi}{3} = 2 \cdot \frac{1}{2} - \frac{1}{2} = \frac{1}{2}$$

f)

$$\frac{\sin \frac{\pi}{4}}{\cos \frac{\pi}{4}} = \frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}} = 1$$

Esercizio 2 – Angoli associati (8 pt)

a)

$$\sin \frac{5\pi}{6} = \sin \left(\pi - \frac{\pi}{6} \right) = \frac{1}{2}$$

b)

$$\cos \frac{5\pi}{6} = -\cos \frac{\pi}{6} = -\frac{\sqrt{3}}{2}$$

c)

$$\tan \frac{2\pi}{3} = -\tan \frac{\pi}{3} = -\sqrt{3}$$

d)

$$\sin \frac{7\pi}{6} = -\sin \frac{\pi}{6} = -\frac{1}{2}$$

e)

$$\cos \frac{4\pi}{3} = -\cos \frac{\pi}{3} = -\frac{1}{2}$$

f)

$$\tan \frac{5\pi}{4} = \tan \frac{\pi}{4} = 1$$

g)

$$\sin \left(-\frac{\pi}{3}\right) = -\sin \frac{\pi}{3} = -\frac{\sqrt{3}}{2}$$

h)

$$\cos \left(-\frac{2\pi}{3}\right) = \cos \frac{2\pi}{3} = -\frac{1}{2}$$

Esercizio 3 – Formule di addizione e sottrazione (10 pt)

a)

$$\begin{aligned}\sin \left(\frac{\pi}{4} + \frac{\pi}{6}\right) &= \sin \frac{\pi}{4} \cos \frac{\pi}{6} + \cos \frac{\pi}{4} \sin \frac{\pi}{6} \\ &= \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}\end{aligned}$$

b)

$$\cos \left(\frac{\pi}{3} - \frac{\pi}{6}\right) = \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$$

c)

$$\begin{aligned}\tan(\alpha + \beta) &= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} \\ &= \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}} = 2 + \sqrt{3}\end{aligned}$$

d)

$$\sin 75^\circ = \sin(45^\circ + 30^\circ) = \frac{\sqrt{6} + \sqrt{2}}{4}$$

e)

$$\cos 15^\circ = \cos(45^\circ - 30^\circ) = \frac{\sqrt{6} + \sqrt{2}}{4}$$

f)

$$\tan 15^\circ = \tan(45^\circ - 30^\circ) = 2 - \sqrt{3}$$

g)

$$\begin{aligned}\cos 105^\circ &= \cos(60^\circ + 45^\circ) \\ &= \frac{1}{2} \cdot \frac{\sqrt{2}}{2} - \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{2} - \sqrt{6}}{4}\end{aligned}$$

h)

$$\sin \frac{5\pi}{12} = \sin \left(\frac{\pi}{4} + \frac{\pi}{6} \right) = \frac{\sqrt{6} + \sqrt{2}}{4}$$

Esercizio 4 – Espressioni (4 pt)

a)

Si riconosce la formula del seno della somma.

$$= \sin \left(\frac{\pi}{6} + \frac{\pi}{3} \right) = \sin \frac{\pi}{2} = 1$$

b)

Formula del coseno della differenza.

$$= \cos \left(\frac{\pi}{4} - \frac{\pi}{6} \right) = \cos \frac{\pi}{12} = \frac{\sqrt{6} + \sqrt{2}}{4}$$

c)

$$\begin{aligned}\sin \left(\frac{\pi}{3} + \frac{\pi}{6} \right) &= \sin \frac{\pi}{2} = 1 \\ \frac{1}{\cos \frac{\pi}{6}} &= \frac{1}{\frac{\sqrt{3}}{2}} = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}\end{aligned}$$

Esercizio 5 (2 pt)

$$\sin 75^\circ \cos 15^\circ + \cos 75^\circ \sin 15^\circ$$

Si riconosce la formula

$$\sin(\alpha + \beta)$$

con

$$\alpha = 75^\circ, \quad \beta = 15^\circ.$$

Quindi

$$= \sin 90^\circ = 1.$$

Risposte finali

Esercizio	Risposta
1a	$\frac{1}{2}$
1b	$\frac{1}{2}$
1c	1
1d	$\frac{3}{4}$
1e	$\frac{1}{2}$
1f	1
2a	$\frac{1}{2}$
2b	$-\frac{\sqrt{3}}{2}$
2c	$-\sqrt{3}$
2d	$-\frac{1}{2}$
2e	$-\frac{1}{2}$
2f	1
2g	$-\frac{\sqrt{3}}{2}$
2h	$-\frac{1}{2}$
3a	$\frac{\sqrt{6}+\sqrt{2}}{4}$
3b	$\frac{\sqrt{3}}{2}$
3c	$2 + \sqrt{3}$
3d	$\frac{\sqrt{6}+\sqrt{2}}{4}$
3e	$\frac{\sqrt{6}+\sqrt{2}}{4}$
3f	$2 - \sqrt{3}$
3g	$\frac{\sqrt{2}-\sqrt{6}}{4}$
3h	$\frac{\sqrt{6}+\sqrt{2}}{4}$
4a	1
4b	$\frac{\sqrt{6}+\sqrt{2}}{4}$
4c	$\frac{2\sqrt{3}}{3}$
5	1